

MERN Stack Interview Questions

Full Stack Developer — Complete Question Bank

Aman Singh · github.com/Aman-singh98

1. Database & MongoDB

1. What is a Relational Database?
2. What is a Non-Relational (NoSQL) Database?
3. What is the difference between SQL and NoSQL databases?
4. What is MongoDB?
5. How does MongoDB store data?
6. What are the advantages of MongoDB?
7. What is an Aggregation Pipeline in MongoDB?
8. What kind of problems can be solved using the Aggregation Pipeline?
9. What are the commonly used stages in an Aggregation Pipeline? (\$match, \$group, \$project, \$sort, \$lookup, \$limit)
10. What is a schema in Mongoose?
11. What is a model in Mongoose?
12. What are indexes in MongoDB?
13. What is normalization and denormalization?
14. What are ACID properties?
15. What is connection pooling?
16. What is MySQL? Difference between SQL and MongoDB?
17. What is ORM? What are Sequelize and Prisma?
18. What are SQL joins? (INNER, LEFT, RIGHT, FULL)
19. What are primary keys and foreign keys?
20. What are parameterized queries? How do you prevent SQL injection?

2. Node.js

1. What is Node.js?
2. Is Node.js a language or a runtime?
3. How does Node.js work internally?
4. Why is Node.js fast?
5. What is the V8 engine?
6. Is Node.js single-threaded? How does it handle multiple requests?
7. What is the Event Loop in Node.js?
8. What is non-blocking I/O? Difference between blocking and non-blocking I/O?
9. What is event-driven architecture?
10. What are Node.js modules? Types of modules?
11. What is require() vs import()? CommonJS vs ES Modules?
12. How do you create an HTTP server in Node.js?
13. What is the fs module? How do you read and write files?
14. What are streams and buffers?
15. What is clustering?
16. What is nodemon?
17. What is dotenv? What is process.env? What is NODE_ENV?
18. What is package.json and package-lock.json?
19. What are dependencies vs devDependencies?
20. What is npm? What is Yarn? Difference between them?

3. Express.js

1. What is Express.js? Why use it? Difference from Node.js?

2. What is routing in Express.js?
3. What is Middleware in Express.js?
4. What are the types of middleware? (application, route, error, built-in)
5. What are the use cases of Middleware?
6. What is the next() function in Middleware?
7. What happens if next() is not called?
8. What is app.use()? Difference between app.use() and app.get()?
9. What is express.json()? What is express.urlencoded()?
10. What is Express Router?
11. What is req object? What is res object?
12. Difference between req.params and req.query?
13. What is error-handling middleware? How do you handle errors in Express?
14. How does the request-response flow work in Express?
15. What is MVC Architecture? Explain Model, View, Controller, Routes.
16. What is the responsibility of Route, Controller, and Model layers?
17. What is the request flow in MVC Architecture?
18. What is CORS? What is Same-Origin Policy?
19. How do you protect routes with authentication in Express.js?
20. How do you build a REST API in Node.js/Express.js?

4. REST API & HTTP

1. What is HTTP? What is HTTPS? Difference between them?
2. What is a REST API? What is an API? Difference between API and REST API?
3. What are HTTP methods? (GET, POST, PUT, PATCH, DELETE)
4. Difference between PUT and PATCH?
5. What is idempotency?
6. What are HTTP status codes?
7. Explain 200, 201, 204, 400, 401, 403, 404, 409, 500.
8. Which status code for duplicate entries? Invalid JWT? Unauthorized? Forbidden?
9. What are request and response headers?
10. Difference between path parameters and query parameters?
11. What is Pagination? Why do we use it?
12. How do you implement Pagination in Node.js?
13. Difference between Offset Pagination and Cursor Pagination?
14. Why is skip() not recommended for large datasets?
15. What is API versioning?
16. What is API validation? How do you validate user input?
17. What is filtering, sorting, and searching in APIs?
18. What is Exception Handling? How do you handle exceptions in Node.js?
19. What is centralized error handling in Express.js?
20. Explain the HTTP request-response lifecycle.

5. Authentication & Security

1. What is JWT (JSON Web Token)?
2. What is the structure of a JWT? (Header, Payload, Signature)
3. How does JWT work?
4. What are the types of JWT tokens?
5. What is the difference between an Access Token and a Refresh Token?
6. What is the expiry time of an Access Token and a Refresh Token?
7. Why do we use both Access Tokens and Refresh Tokens?
8. Where do you store JWT tokens? (localStorage vs httpOnly cookie)

9. How do you verify a JWT token? How do you protect routes using JWT?
10. Difference between JWT and Session Authentication?
11. What is Authentication vs Authorization?
12. What is hashing? What is bcrypt? What is salting?
13. Difference between hashing, encryption, and encoding?
14. What is SQL Injection? How do you prevent it?
15. What is XSS? What is CSRF?
16. What is Helmet? What is Rate Limiting?

6. JavaScript

1. What is JavaScript? What are the data types?
2. Difference between primitive and non-primitive data types?
3. Difference between var, let, and const?
4. What is Hoisting?
5. What is the Temporal Dead Zone (TDZ)?
6. What is Scope in JavaScript? What is Lexical Scope?
7. What is a Closure? Real-world use cases?
8. What is the Event Loop? Call Stack, Web APIs, Callback Queue, Microtask Queue?
9. What is a Promise? What are the states of a Promise?
10. Promise vs Callback? What is Callback Hell?
11. Promise.all() vs Promise.allSettled() vs Promise.race() vs Promise.any()?
12. What are async and await? Can we use await without async?
13. What is this keyword in JavaScript?
14. Difference between == and ===?
15. Difference between null and undefined?
16. What are call(), apply(), and bind()?
17. What is Currying? What is Infinite Currying?
18. What is Recursion?
19. What is Debouncing? What is Throttling?
20. What is Memoization?
21. Difference between map(), filter(), reduce(), and forEach()?
22. Difference between find() and filter()?
23. Difference between slice() and splice()?
24. What is destructuring? Spread and rest operators?
25. What are template literals? Optional chaining? Nullish coalescing (??)?
26. Difference between Set and Map?
27. How do you remove duplicate elements from an array?
28. What is prototype? What is prototype inheritance?
29. What is event bubbling, capturing, and delegation?
30. What is JSON? JSON.parse() vs JSON.stringify()?
31. What is fetch()? Fetch API vs Axios?
32. What is exception handling? try-catch-finally?
33. What is a higher-order function?
34. What are ES6 features?
35. How do you optimize JavaScript performance?

7. React

1. What is React?
2. What is JSX?
3. What is Virtual DOM? What is the DOM? Difference between DOM and Virtual DOM?
4. What is Reconciliation?

5. What are React components? Functional vs Class components?
6. What is the React Component Lifecycle? Lifecycle Methods?
7. How do Lifecycle Methods work in Functional Components using Hooks?
8. What are Props? What is State? Difference between Props and State?
9. What is useState()?
10. What is useEffect()? Explain the dependency array.
11. What is useRef()?
12. What is useMemo()?
13. What is useCallback()? Difference between useMemo and useCallback?
14. What is React.memo()? How does it improve performance?
15. What is Context API?
16. What is Redux? What is Redux Toolkit?
17. Difference between Context API and Redux? When to use which?
18. What is Zustand? Difference between Zustand and Redux? When to use Zustand?
19. What is Prop Drilling?
20. What are Controlled and Uncontrolled Components?
21. What are React Hooks? Custom Hooks?
22. What is React Router? Difference between Link and anchor tag?
23. What are Dynamic Routes?
24. What are Higher Order Components (HOC)?
25. What is Lazy Loading? What is Code Splitting? What is Suspense?
26. What are React Fragments?
27. What are Keys in React?
28. How do you call APIs in React? Handle loading and error states?
29. How do you optimize a React application?
30. What are Controlled and Uncontrolled Components?

8. HTML & CSS

1. What is HTML? What is HTML5?
2. What are semantic HTML tags?
3. Difference between id and class?
4. Difference between div and span?
5. What are block-level and inline elements?
6. What are forms in HTML? Difference between GET and POST in forms?
7. What are localStorage, sessionStorage, and Cookies?
8. What are data attributes? What are meta tags?
9. What is the DOCTYPE declaration?
10. What are ARIA accessibility attributes?
11. What is SEO-friendly HTML?
12. What is CSS? Different ways to apply CSS?
13. What is CSS specificity?
14. What is the CSS Box Model?
15. Difference between margin and padding?
16. Difference between display:none and visibility:hidden?
17. Difference between relative, absolute, fixed, and sticky positioning?
18. What is z-index?
19. What is Flexbox? What is CSS Grid? Difference between them?
20. What are media queries? What is responsive design? What are breakpoints?
21. Difference between px, %, em, rem, vh, and vw?
22. What are pseudo-classes and pseudo-elements?
23. What are CSS variables?
24. Difference between transition and animation?

25. How do you center a div?

9. Next.js & SEO

1. What is Next.js? Which type of code can we write in Next.js?
2. What is SSR, SSG, and ISR?
3. What is the App Router in Next.js 15?
4. What are Server Components vs Client Components?
5. Can we build REST APIs in Next.js? Is it good practice?
6. When should you choose Next.js API Routes instead of Express.js?
7. Why is Next.js better for SEO than React?
8. What is SEO? Why is SEO important?
9. What is Technical SEO?
10. What is a CDN and why is it used?

10. File Uploads & Real-time (Socket.IO)

1. How do you upload files in Express?
2. What is Multer? How do you store uploaded files? Validate file types?
3. What is Cloudinary? How do you integrate it with Node.js?
4. What is Socket.IO? Why use it?
5. How does Socket.IO work? Difference between Socket.IO and WebSocket?
6. What are Rooms and Namespaces in Socket.IO?
7. How do you implement a chat application?
8. How do you authenticate Socket.IO connections?
9. How do you scale Socket.IO?

11. Git & GitHub

1. What is Git? What is GitHub? Difference between them?
2. What is GitLab? Difference between GitHub and GitLab?
3. What is a Repository? What is a Commit? What is a Branch?
4. What is Git Merge? What is Git Rebase? Difference between them?
5. What is a Merge Conflict? How do you resolve it?
6. What is Git Stash?
7. What is a Pull Request? What is a Merge Request?
8. What is Git Flow?
9. Difference between git fetch and git pull?
10. What are the common Git commands?
11. Explain your Git workflow.

12. AWS & Deployment

1. What is AWS?
2. What is EC2? What is S3? Difference between EC2 and S3?
3. What is RDS? What is DynamoDB?
4. What is IAM? What is Route 53?
5. What is CloudFront? What is CloudWatch?
6. What is VPC? What is Security Group? Difference between Security Group and NACL?
7. What is Auto Scaling?
8. What is Elastic Load Balancer?
9. What is Lambda? Difference between EC2 and Lambda?
10. What is API Gateway?
11. What is a VPS? Difference between VPS and Shared Hosting? VPS vs EC2?
12. What is Nginx? What is a Reverse Proxy?

13. What is PM2?
14. What is SSH? What are ports 22, 80, and 443?
15. Common ports for Node.js, MySQL, MongoDB, and Redis?
16. How do you deploy a React app on AWS?
17. How do you deploy a Node.js app on AWS?
18. What is CI/CD? What is GitHub Actions?

13. System Design (Basic)

1. Explain your project architecture.
2. How do you design a scalable REST API?
3. How do you optimize database queries?
4. How do you handle concurrent users?
5. How does a browser communicate with a server? What happens when you type a URL?
6. What is the request-response lifecycle?
7. What is a proxy server? What is a reverse proxy?
8. What is caching? Why use Redis for caching?
9. What is Load Balancing? Horizontal vs Vertical Scaling?
10. What are Read Replicas?
11. What are Microservices? Why are they used?
12. What is Monolithic Architecture?
13. What is Kafka or RabbitMQ? Why are they used?
14. What is Adaptive Bitrate Streaming?
15. How does Hotstar handle millions of concurrent users?
16. What is Redis? Redis vs TanStack Query?
17. How do you optimize backend performance?

14. Generative AI & LLMs

1. What is Generative AI (GenAI)?
2. What is an LLM (Large Language Model)?
3. What is Prompt Engineering?
4. What is RAG (Retrieval-Augmented Generation)?
5. How do you integrate GenAI into a Node.js application?

15. TypeScript & Dev Tools

1. What is TypeScript? Difference between TypeScript and JavaScript?
2. What are Interfaces in TypeScript?
3. What are Enums in TypeScript?
4. What are Generics in TypeScript?
5. TypeScript with React — typing props, state, events, refs?
6. TypeScript with Express — typed req/res, custom Request interface?
7. What is Webpack? What is Vite? Difference between them?
8. What is npm? What is Yarn? Difference?
9. What is Figma?
10. What is Agile? What is Scrum? What is JIRA?

16. Coding Questions

1. Create a simple HTTP server in Node.js.
2. Create an Express server with basic routing.
3. Build a complete CRUD API.
4. Build a Signup API.
5. Build a Login API.

6. Build JWT Authentication (access token + refresh token).
7. Build an Authentication Middleware.
8. Connect Express with MySQL using Sequelize.
9. Connect Express with MongoDB using Mongoose.
10. Build a Pagination API.
11. Build a Search API.
12. Upload a file using Multer.
13. Implement a Promise-based delay function.
14. Write a Fetch API GET request.
15. Write a Fetch API POST request.
16. Implement debouncing from scratch.
17. Implement throttling from scratch.
18. Implement memoization from scratch.
19. Remove duplicate elements from an array (multiple approaches).
20. Write map(), filter(), reduce() from scratch.
21. Build a complete REST API with auth, CRUD, pagination, and file upload.

17. HR & Project-Based Questions

1. Tell me about yourself.
2. Explain your current project.
3. What are your responsibilities?
4. What challenges have you faced and how did you solve them?
5. How do you debug production issues?
6. Explain a difficult bug you fixed.
7. Explain your deployment process.
8. Explain your Git workflow.
9. Why are you looking for a job change?
10. Why should we hire you?
11. What are your strengths and weaknesses?
12. Where do you see yourself in 5 years?
13. What is your notice period?
14. What are your salary expectations?